

Cloud Native Streaming Platform Netflix-inspired

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1 Executive Summary

Netflix is a streaming platform that provides this service for subscriptions. Whose vision is entertaining customers through series and movies. This content can be accessed around the world by millions of customers those who want to watch their favorites shows at any time. For that reason is an application demands great availability, good performance with heavy content such as multimedia and personalized experience with assertive recommendations.

2 System Description

Business problem, actor catalog, use case definitions, and system boundary diagram

2.1 Business Problem

Netflix solves the problem of providing a large amount of audiovisual content to millions of users through the internet while giving each user a convenient and personalized experience.

Users want to watch movies and series whenever and wherever they want without having to download the entire content beforehand. At the same time, Netflix needs to help users find something they want to watch among a very large catalog.

The platform provide another services that not will be consider in this project like: Option for subtitles, interactive experiences or content preview.

2.2 Actors

The application has 4 users represented by a person, each of this person has a particular function and use the application for different purposes. Also there are some system actor that provide services for the application.

Actor	Role	Frequency of interaction	Expectation
Customer	A person who has a monthly membership plan with the platform.	High	Be able to watch series and movies without interruptions and receive relevant recommendations.
Operational user	A member of the operations team responsible for managing content within the system.	Medium	Be able to manage all the content without problems.
Data analyst	A member of the Data & Insights area who can access data and statistics.	Medium	Have access to useful data with clear statistics.
Platform administrator	Is responsible for managing users and the business model.	Low	Be able to manage the users of the system and membership configurations.
Media Convert service	It allows videos to be sent in small segments.	High	The user does not need to download the entire video; it is stored in fragments in the device buffer.
CDN service	Servers that help reduce latency by delivering content from the nearest server.	High	Ensure rapid response to high volumes of requests.
Storage	Storage for multimedia content and metadata, useful for performing queries.	High	Persist all data over time.

Table 1: Actors, roles, interaction frequency, and expectations of the Netflix platform.

2.3 Main Use Cases

Subsection example 2

2.4 System Boundaries

Subsection example 2

3 Functional Requirements

This requirements specify what application does. There are 11 functional requirements divided in domain

4 Non-Functional Requirements

Numbered catalog of measurable non-functional requirements per category

5 Workload Model

Structured table with all workload parameters and justification for each value

References

- [1] Rummery, G. A., & Niranjana, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.